



MyEyes

Garments Detection and Classification using Region based - CNNs

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Abstract— Although technological developments in the last few years have been exciting, there are still many gaps when it comes to helping those who are blind. How we dress reflects the person we are in the world. Every day we face a dress code for work, going out to dinner, in a musical festival, among others. Something that for most of the population is a simple and stress-free task can be very hard for those who are blind. Previous work shows that for the blind community not knowing garment types, colors, and patterns is a demanding challenge [1]. Nowadays, there are some apps to assist the blind in planning what to wear and combining clothes despite not having been developed with them in mind [2]. When it comes to the detection of garments, we can find projects that detect the garment using Conditional Random Field [3], yet it shows some problems with clothes used on the same body part. Another project that can be found uses a deformable semantic locality-preserving network demonstrating good results regarding the corners of the pieces of clothing [4]. Focused on blind people we can find projects that identify the colors and patterns of clothes [5]. Additionally, MyEyes has previously undertaken efforts to assist blind individuals, including the development of combination systems [6-9]. However, until now, no previous endeavors have specifically targeted the automatic identification of characteristics in clothing items. Therefore, the project presented in this study makes a significant contribution to this field, enabling the automatic extraction of features for the identification of clothing items using artificial intelligence algorithms. This is achieved by extracting garments from images with varying backgrounds and cataloging them in a database to augment the effectiveness of the algorithms in identifying and extracting the clothes' characteristics [10]. The resulting dataset comprises polygonal segmentation for locating clothes in 1,120 images across eight garment categories (classes), including dresses, jackets, pants, polos, shirts, shoes, shorts, and t-shirts. By using Region Based Convolution Neural Networks (R-CNNs), it was possible to detect objects, including garments. To train and test the garment recognition network, we utilized the dataset developed in this study, which was split into training and validation images with ratios of 0.8 and 0.2, respectively. For the eight classes of garments detected, label results are above 77% regarding average precision for all categories. In the experiments performed we evaluated the influence of a different number of training iterations, data augmentation, and preprocessing, analyzing the impact on the results. To streamline and enable future research efforts, the collected data is stored in a centralized database.

Keywords— *Blind people; Garments recognition; Machine Learning; Object Detection.*

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TOPIC

- 3) Intelligent Systems
 - a. Identification systems.

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